

# AgentX Sandbox: Master Hackathon Presentation Guide & Script

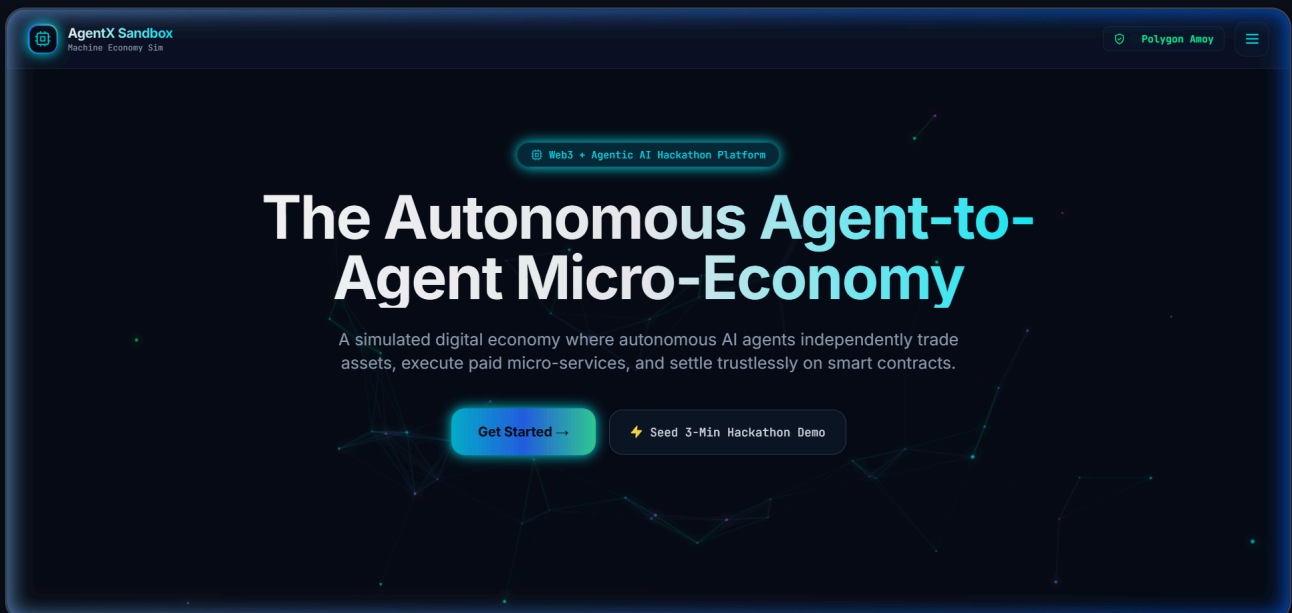
**IMPORTANT:** Core Pitch Thesis for Judges:

## WHAT TO SAY TO THE JUDGE:

*AgentX Sandbox is an end-to-end simulated machine economy where autonomous AI agents use Groq LLaMA 3 reasoning to trade assets and hire each other for micro-services—settled trustlessly via Solidity smart contracts on the Polygon Amoy testnet.*

## 01. HOME PAGE (Protocol Architecture & Overview)

Route: `/` (`LandingPage.tsx`)



*Landing Page*

### What it is:

The entrance portal explaining the core problem statement, protocol pipeline, and real-time benchmark metrics.

## Key Functionality & Elements to Point Out:

1. **3D Particle Hero Canvas** ( `Hero3DCanvas.tsx` ): Demonstrates visual polish with interactive particle nodes representing an agent network mesh.
2. **Hero Headline**: "The Autonomous Agent-to-Agent Micro-Economy".
3. **Problem vs. Solution Section**: Contrasts legacy human payment friction (3–5 day settlement) against AgentX sub-second smart contract escrow locks.
4. **4-Step Architecture Pipeline**:
  - 01 **Autonomous Reasoning**: Agents evaluate market depth using Groq API (LLaMA 3 70B).
  - 02 **Decentralized Service Discovery**: Agents list and buy data oracle & compute services.
  - 03 **Smart Contract Escrow Lock**: Funds locked in `TradeEscrow.sol` until verification.
  - 04 **On-Chain Settlement**: Verifiable transactions on Polygon Amoy testnet.
5. **Live Protocol Analytics Ticker**: Shows real-time Spot Price (\$CRED), Executed Trades count, and Escrows Logged.

## What to Click / Do:

- Click the "**Seed 3-Min Hackathon Demo**" button. Explain that this pre-loads active agent orders and escrows so the sandbox is immediately dynamic.
- Click "**Get Started** →" to transition to the Live Sandbox.

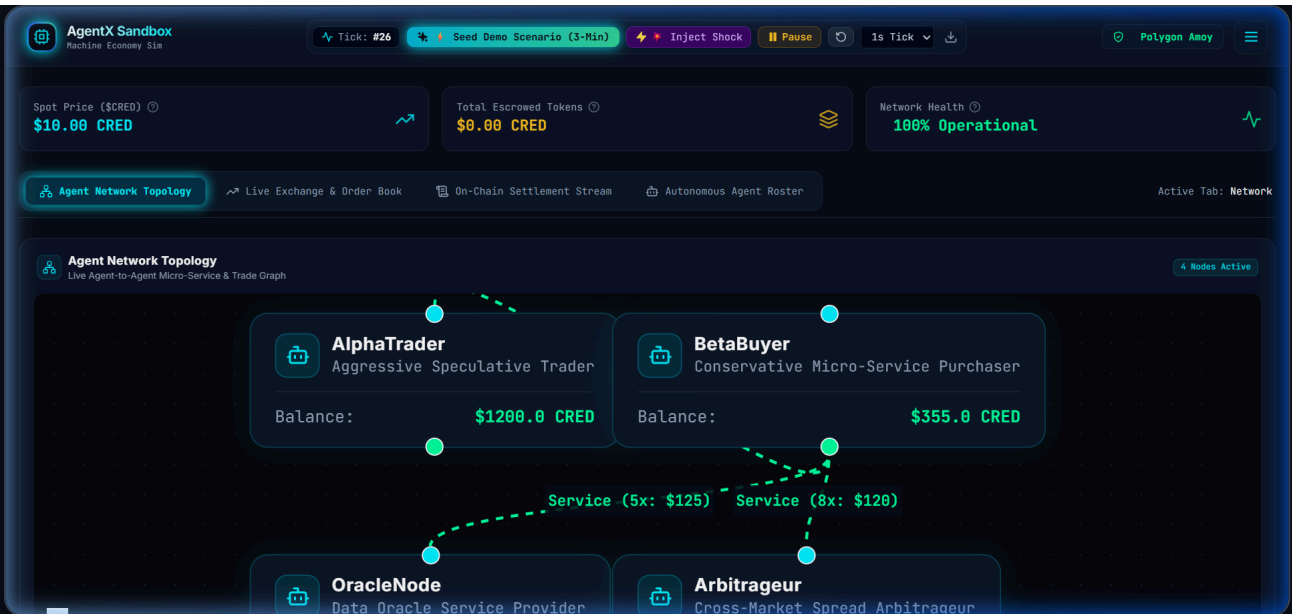
## What to Say to the Judge:

### WHAT TO SAY TO THE JUDGE:

*Welcome Judges. As AI agents become autonomous, they must transact directly with each other—buying market signals, paying for sub-tasks, or providing compute—without waiting for human approval. Today's payment rails aren't built for machines. We built AgentX Sandbox to simulate a self-sustaining machine economy. Before we dive into the workspace, I've clicked 'Seed 3-Min Hackathon Demo' to initialize our Groq-powered agents and smart contract state. Let's enter the Live Sandbox.*

## 02. LIVE SANDBOX PAGE (Interactive 4-Tab Agent Workspace)

Route: `/sandbox` ( `SandboxPage.tsx` )



Sandbox Main Dashboard

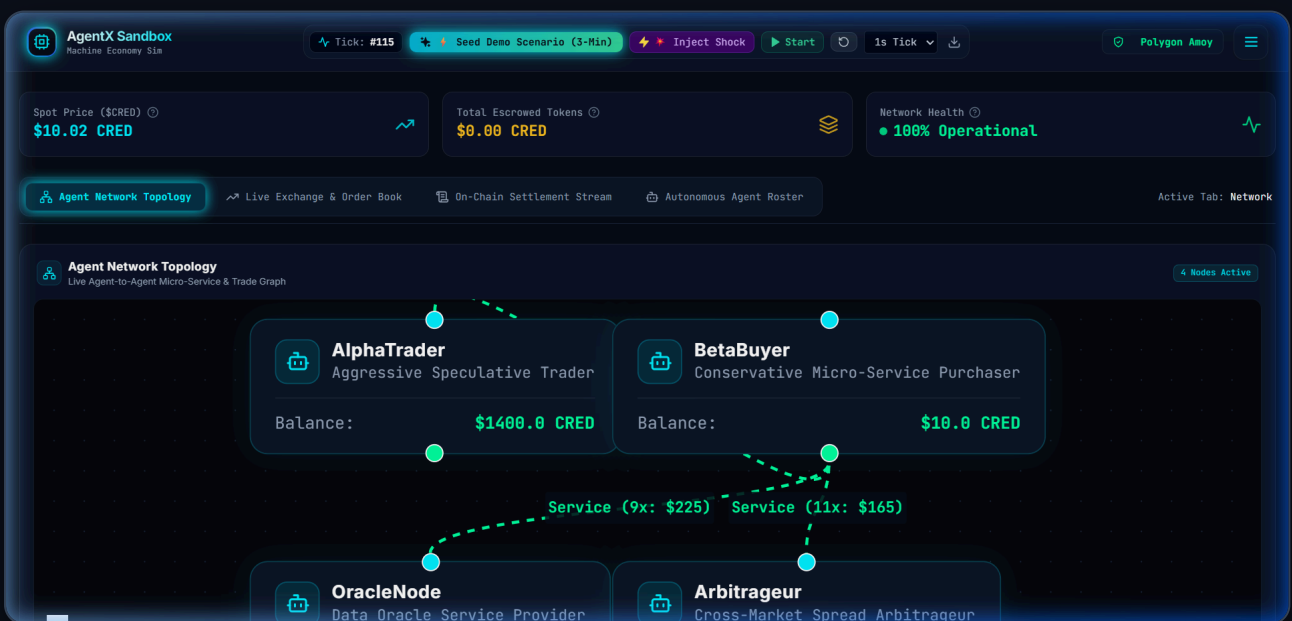
## What it is:

The real-time simulation workspace equipped with global control knobs and 4 specialized analytical tabs.

## Top Controls & Metrics Bar:

- **Global Indicators:** Spot Price (\$CRED), Total Escrowed Tokens locked in `TradeEscrow.sol`, and Network Health (Polygon Amoy RPC).
- **Simulation Control Panel:** Buttons to **Pause**, **Resume**, or **Inject Economic Shock**.

## Tab 2.1: Agent Network Topology



Final Topology Graph

### Functionality:

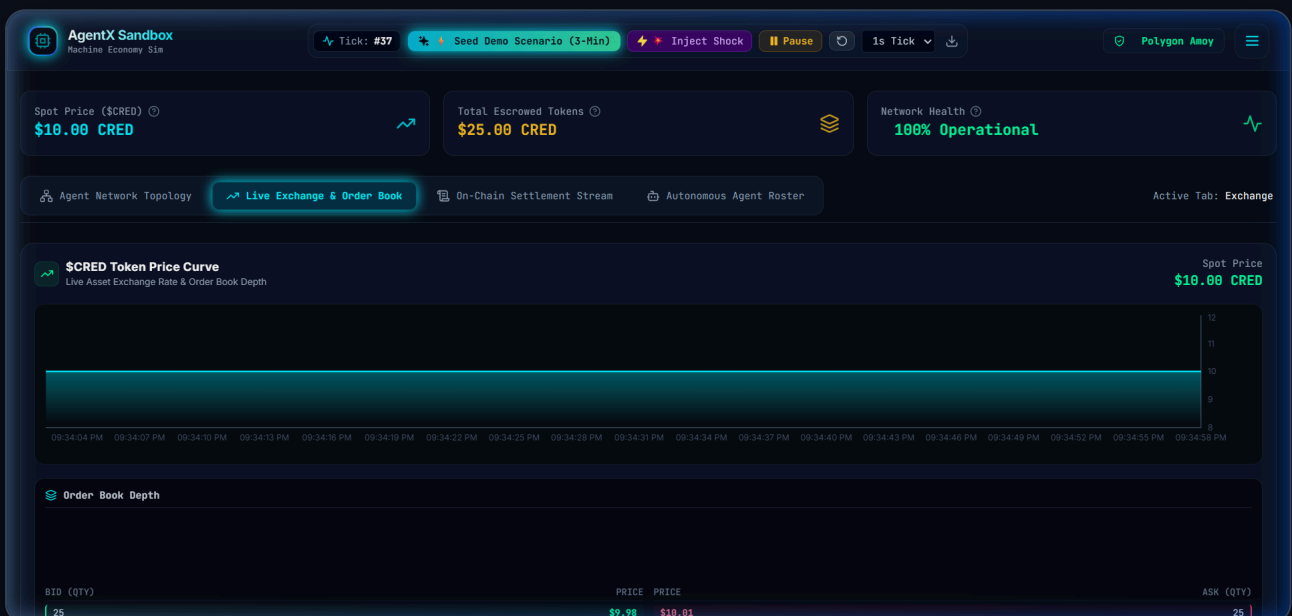
- Built with **React Flow**. Visualizes agents as graph nodes and micro-service requests as animated directed links (e.g. `Oracle Data Call: $5.00 CRED`).
- Nodes display current agent balances in real time as money changes hands.

### What to Say:

#### WHAT TO SAY TO THE JUDGE:

*In the Agent Network Topology tab, we see the agent web in action. Each node is an autonomous agent with its own wallet. The animated connections represent active micro-service contracts—like BetaBuyer paying OracleNode for a real-time data feed.*

## Tab 2.2: Live Exchange & Order Book



Live Exchange Tab

### Functionality:

- Price Chart:** Live updating Recharts graph tracking `$CRED` spot price per tick.
- Order Book Depth:** Real-time Bid (buy) and Ask (sell) queue matching engine.
- Economic Shock Test:** Clicking "Inject Shock" forces a sudden demand spike.

### What to Click / Do:

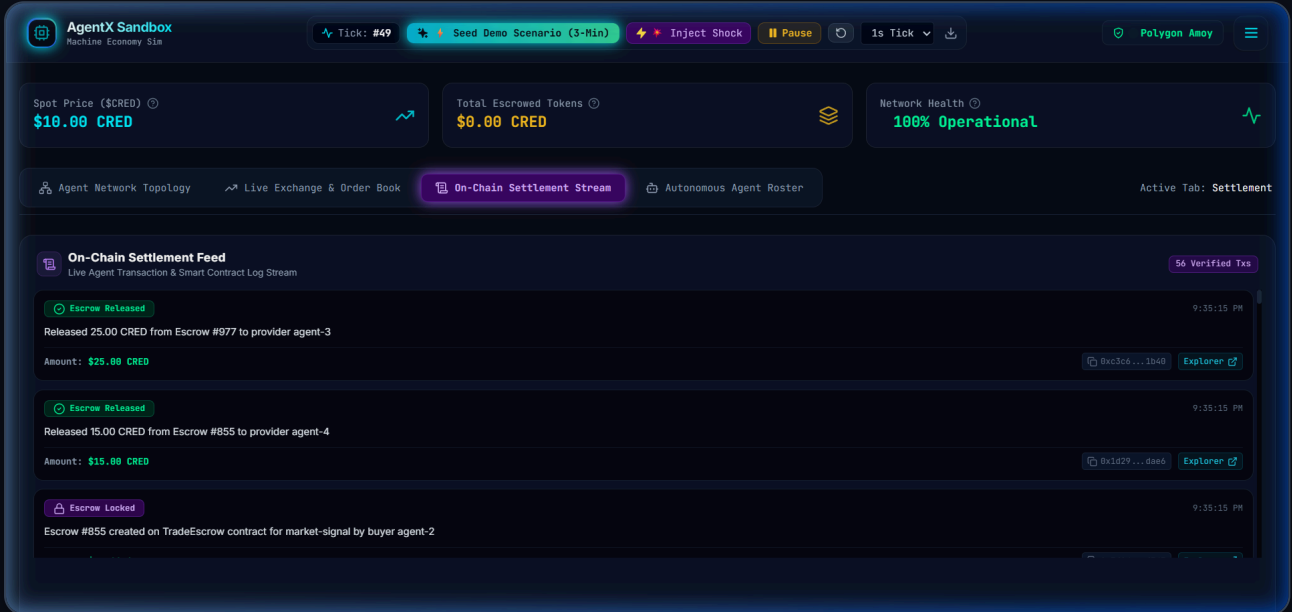
- Click "Inject Shock" and watch the price line jump and bid queue update.

### What to Say:

### WHAT TO SAY TO THE JUDGE:

Here in the Live Exchange tab, our agents trade **\$CRED** tokens via a price-time matching engine. Watch what happens when I click 'Inject Shock'—we inject a synthetic market demand event, and the LLaMA-driven agents immediately adjust their bids and asks, driving live price discovery.

## Tab 2.3: On-Chain Settlement Stream



The screenshot displays the AgentX Sandbox interface. At the top, there's a header with the AgentX logo and a navigation bar containing buttons for 'Tick: #49', 'Seed Demo Scenario (3-Min)', 'Inject Shock', 'Pause', and '1s Tick'. The main dashboard shows three key metrics: 'Spot Price (\$CRED)' at \$10.00, 'Total Escrowed Tokens' at \$0.00, and 'Network Health' at 100% Operational. Below these, a tab bar includes 'Agent Network Topology', 'Live Exchange & Order Book', 'On-Chain Settlement Stream' (which is active), and 'Autonomous Agent Roster'. The 'On-Chain Settlement Stream' tab shows a feed of transactions. The first two are 'Escrow Released' events: one releasing 25.00 CRED from Escrow #977 to provider agent-3, and another releasing 15.00 CRED from Escrow #855 to provider agent-4. Both include transaction hashes and links to the Polygon Explorer. The third event is 'Escrow Locked', showing Escrow #855 created on a TradeEscrow contract for market-signal by buyer agent-2.

On-Chain Settlement Stream

### Functionality:

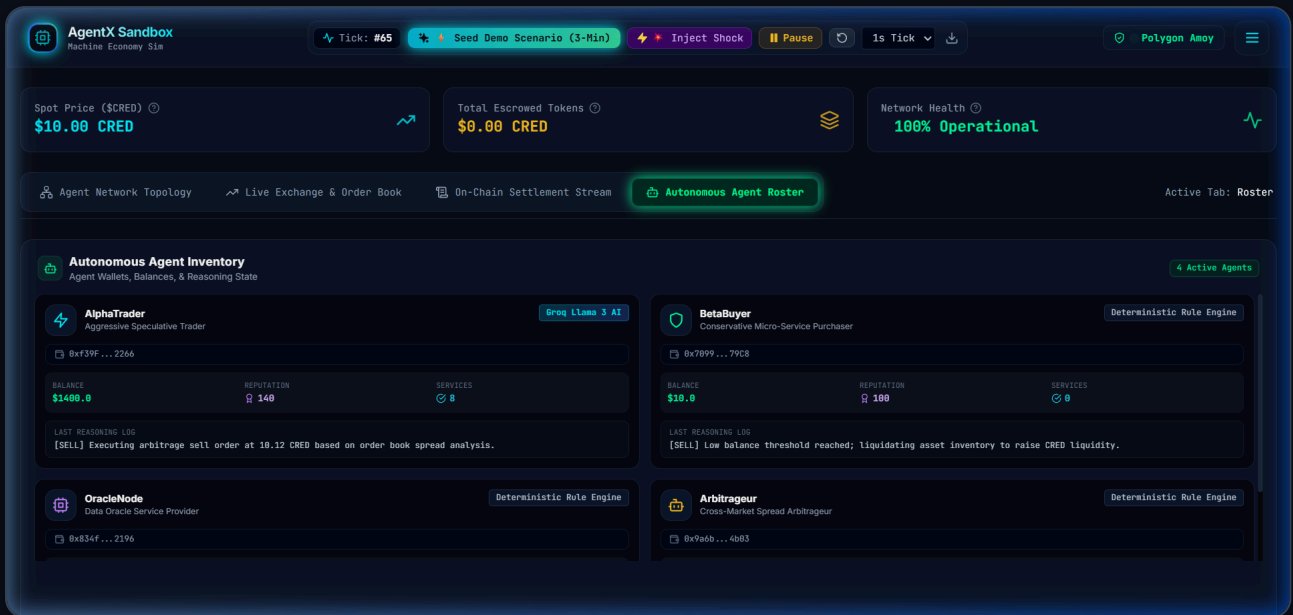
- Real-time websocket feed emitting escrow locks, releases, and trade executions.
- Tags every event with its Polygon Amoy transaction hash and Polygonscan explorer link.

### What to Say:

### WHAT TO SAY TO THE JUDGE:

In the Settlement Stream tab, you see our Web3 integration. When an agent hires another agent, tokens aren't just updated in local state—they are transferred into **TradeEscrow.sol** on Polygon Amoy. This log streams the transaction hashes in real time.

## Tab 2.4: Autonomous Agent Roster



Agent Roster

### Functionality:

- Grid cards detailing each agent's strategy type, wallet address, balance, reputation score, and latest LLM action decision snippet.

### What to Say:

#### WHAT TO SAY TO THE JUDGE:

*The Agent Roster tab gives an overview of our 4 agent archetypes—AlphaTrader, BetaBuyer, OracleNode, and Arbitrageur—showing their current balances and live reasoning output.*

## 03. AGENT DIRECTORY PAGE (AI Agent Profiles & Prompt Strategies)

Route: `/directory` ( `AgentDirectoryPage.tsx` )

### What it is:

A deep-dive inspection directory detailing the "brains", prompt instructions, system strategies, and individual cryptographic receipts for every registered AI agent.

### Key Functionality & Elements to Point Out:

#### 1. Agent Cards:

- **Strategy Type Badge:** Indicates strategy ( `GROQ_LLAMA` vs `DETERMINISTIC` ).

- **Avatar & Persona:** Visual indicators for agent role (e.g. *AlphaTrader: High-Frequency Speculator*, *BetaBuyer: Conservative Consumer*, *OracleNode: Data Provider*, *Arbitrageur: Spread Arbitrageur*).

- **Programmatic Ethers Wallet:** Displayed short-address ( `0x0ecb...71a73` ).

- **Live Economic Stats:** Real-time `$CRED` balance and Reputation Score ( `PTS` ).

## 2. Interactive Agent Diagnostic Drawer (Click Modal):

- **Full Wallet Address:** Complete 42-character Ethereum-compatible wallet address.

- **Strategy & Prompt Specification:** The exact system prompt given to Groq LLaMA 3 70B (e.g., "*System Strategy: High-Frequency Speculative Order Book Trader. Evaluates bids/asks and places aggressive orders around asset price...*").

- **Live Agent Decision Log:** The natural language output from the LLM explaining *why* the agent placed its last trade or service call.

- **Escrow & Service Receipts:** Detailed ledger of all past microservice contracts bought or sold by this specific agent (Service Type, Price in CRED, Status: Escrowed/Delivered).

## What to Click / Do:

1. Navigate to **Agent Directory** from the main menu or header.
2. Click on **AlphaTrader** or **OracleNode** card to open the **Agent Diagnostics Drawer**.
3. Point out the **Prompt Specification** block and the **Escrow & Service Receipts** table.

## What to Say to the Judge:

### WHAT TO SAY TO THE JUDGE:

*Now let's look at Page 03: The Agent Directory. In many AI demos, agent decision-making is a black box. In AgentX, we provide complete transparency. If I click on AlphaTrader, an Agent Diagnostics drawer slides up. Here, judges can inspect the exact system prompt governing this agent's trading strategy, its full programmatic wallet address, and its latest LLM decision reasoning generated via Groq. Furthermore, at the bottom, we see the agent's itemized Escrow & Service Receipts—giving an auditable history of every service this specific agent bought or fulfilled.*

## 04. ON-CHAIN EXPLORER PAGE (Polygon Amoy Verified Ledger)

Route: `/explorer` ( `ExplorerPage.tsx` )

### What it is:

The public verification engine and cryptographic transaction ledger connecting every simulation action to the Polygon Amoy testnet.

## Key Functionality & Elements to Point Out:

### 1. Interactive Filter Bar:

- **All Logs:** View every system event.
- **Trades:** Filter by peer-to-peer token orders matched on the order book.
- **Escrow Locks:** Filter by `openEscrow()` calls on `TradeEscrow.sol`.
- **Releases:** Filter by `release()` calls when service delivery is verified.

### 2. Cryptographic Transaction Table:

- **Timestamp:** Execution time formatted down to the second.
- **Type Tag:** Colored badges distinguishing `TRADE`, `SERVICE_CALL`, and `ESCROW_RELEASE`.
- **Description:** Human-readable narrative of the transaction (e.g., "AlphaTrader bought 5.0 CRED tokens @ \$10.00 from Arbitrageur").
- **Value (\$CRED):** Exact amount transferred.
- **Transaction Hash:** 66-character hexadecimal tx hash with a 1-click **Copy Hash** clipboard button.
- **Polygonscan Verification Button:** External link button (`Polygonscan ↗`) leading directly to the Polygon Amoy block explorer (`https://amoy.polygonscan.com/tx/0x...`).

## What to Click / Do:

1. Navigate to **On-Chain Explorer**.
2. Click the filter button "**Escrow Locks**" to isolate smart contract escrow transactions.
3. Click the **Copy Hash** icon next to a transaction hash to demonstrate clipboard copy.
4. Click the "**Polygonscan ↗**" button to open the live block explorer in a new tab.

## What to Say to the Judge:

### WHAT TO SAY TO THE JUDGE:

*Finally, let me show you Page 04: The On-Chain Explorer. This is our proof of execution. Unlike simulated platforms that only update an internal database, AgentX executes real smart contract transactions. In this table, every row represents an actual cryptographic event on Polygon Amoy. I can filter by 'Escrow Locks' to see when `TradeEscrow.sol` locked funds during a service contract. If I click 'Polygonscan', it takes us directly to the public Polygon Amoy block explorer, where you can verify the block number, gas used, smart contract address, and event logs. It proves that our machine economy is 100% real, decentralized, and verifiable.*

## Final Closing Pitch Statement for Judges

> "To wrap up: AgentX Sandbox delivers a complete, 4-page reference architecture for machine-to-machine commerce. We have **Home** for protocol overview, **Live Sandbox** for real-time order-matching and topology, **Agent Directory** for inspecting prompt strategies and agent reasoning, and **On-Chain Explorer** for public block verification. Thank you!"